

Vaishnav Gaikwad

Site Reliability & DevOps Consultant



Summary

Skilled and experienced Site Reliability/DevOps Engineer with a strong focus on Kubernetes and Cloud-Native ecosystem. I have more than four years of hands-on experience in deploying and managing cloud-native applications and infrastructure.

I am an individual with great adaptability, problem-solving, collaboration and communication skills and having attention to detail. I always strive to stay ahead of the curve by continuous learning.

Skills

- **Cloud Technology:** Kubernetes Ecosystem, Openshift Container Platform, Azure Kubernetes Service, Elastic Kubernetes Service, RKE, RKE2, Rancher, Helm, Docker, Podman, Portworx CSI, G42 CSI, containerd, Istio (intermediate), Policy Engine (OPA Gatekeeper), Container Security, Operating System fundamentals, Kubernetes networking, Nginx App Protect.
- **Kubernetes Monitoring and Logging:** Prometheus, Alertmanager, Thanos, Grafana, Fluentd, ELK stack, Nginx Instance Manager.
- **Cloud Platforms:** Microsoft Azure, AWS, Google Cloud(Novice), G42(Novice), VMware vSphere(Intermediate).
- **Databases:** SQL, NoSQL Databases - MongoDB, Redis(Novice).
- **Programming languages:** Bash, Go, Python(Novice), SQL, Previous experience: Java, NodeJS.
- **Infrastructure as Code (IaC):** Skilled in using tools such as Terraform and Helm to define and provision infrastructure resources and application deployments.
- **DevOps Tools:** Git, GitHub Actions, Gitlab, ArgoCD, Fleet, Jira, Confluence, Jfrog Artifactory, Jfrog Xray, Redhat Quay, Palo Alto Prisma, Markdown.

Work Overview

OSS contributions:

- **Kiali:** Kiali is a management console for Istio service mesh. When using Kiali with *kubectl proxy*, the url (window.web_root url) at which Kiali is being served is rewritten to a default url (server.web_root) making the service unavailable to access when we refresh. As a solution to this problem I modified the router to pick up the url which is used to access the service and it prints that url in the window. With this modification when we put Kiali

behind *kubectl proxy* the url is rewritten to the same url being used to access the service.

Issue link: <https://github.com/kiali/kiali/issues/4459>

PR link: <https://github.com/kiali/kiali/pull/4914>

Project(s) experience @ Huco (08/2023 - present)

- **SRE & Devops Consultant:**

- Designed, deployed, and operated secure, air-gapped and public cloud OpenShift and RKE2 clusters, adhering to CIS benchmarks and enterprise hardening best practices.
- Provisioned and managed Azure Kubernetes Service (AKS) clusters with cluster autoscaling, node pools, and spot instances for cost optimization while maintaining performance SLAs across development and production environments. Also implemented Azure Key Vault integration with AKS workloads using CSI drivers and managed identities for secure secret management and certificate rotation.
- Built and maintained end-to-end CI/CD pipelines for containerized applications using GitOps tools and integrated artifact scanning via JFrog Xray.
- Configured and managed enterprise-grade ingress and L7 traffic management solutions including F5 CIS, NGINX Ingress Controller, standalone NGINX, and App Protect.
- Implemented and managed Istio service mesh, Kubernetes network policies, and Prisma Cloud for container runtime security and threat prevention.
- Deployed and operated observability stacks with Prometheus, Grafana, Thanos, Alertmanager, Fluentd, and ELK, ensuring high availability, scalability, and SLO compliance.
- Provisioned dynamic and static storage with G42, vSphere, Portworx CSI drivers; developed backup and DR strategies for production-grade Kubernetes clusters.
- Established RBAC policies, namespace segregation, and cluster-level governance; assisted development teams with app onboarding and operational support.
- Delivered high availability Rancher setups for central management of RKE2 clusters and led DevOps adoption initiatives for clients embracing cloud-native platforms.

Project(s) experience @ Infracloud Technologies (07/2021 - 08/2023)

Suse Rancher (08/2021 - 08/2023)

- SRE working in the Kubernetes ecosystem. Helping Rancher integrate monthly kubernetes patches and minor versions.
- Maintain and upgrade multiple helm charts supported by the customer.
- Help end users and open-source community members using rancher and RKE with their issues related to provisioning and tools like Istio, Gatekeeper and CIS-benchmark.

Projects

- **Rancher Kubernetes Engine (RKE & RKE2) -**
 - Undertook the responsibilities to add support for the new docker versions(23.0) to RKE. With this Rancher, RKE, RKE2 and K3s users were able to provision kubernetes clusters on machines having docker version 23.0.
 - Introduced multiple docker versions to the rancher's docker installation shell scripts which is used to install docker on different OSes.
 - Integrated monthly kubernetes patches and minor versions with RKE and Rancher. This includes analyzing the changelog of each minor and patch version and implementing these changes in RKE respectively.
 - Added support for newer versions of nginx-ingress-controller and calico/canal CNIs to RKE.
 - Removed multiple deprecated legacy features which were tightly coupled in Rancher's code to optimize the codebase.
 - Resolved multiple longstanding issues related to CNIs and k8s conformance test.
- **Feature charts -**
 - Maintaining, upgrading and improving helm charts for Istio, Kiali, Jaeger-tracing, OPA-Gatekeeper and CIS benchmark.
 - Resolved any issues logged by the customers and open source community related to these tools and helm charts.
 - Implemented hull framework which is a comprehensive testing framework for helm charts for the opa-gatekeeper chart.
 - Got hands-on experience with cluster and container security tools like trivy, kube-bench, sonobuoy while working on the cis-benchmark chart.
 - Worked on adding PSA support to these charts after PSPs were removed in k8s 1.25.

Educational Qualification

- **Bachelor of Engineering in Computer Science (2018-2021):**
 - Completed my Bachelor's degree in Computer science from Zeal Institute of Engineering and Research.
- **Diploma in Computer Science (2015-2018):**
 - Completed my Diploma in Computer Science from Government Polytechnic Pune.
- **SSC (2014-2015):**
 - Completed my SSC from Holy Cross Convent High School Akola in 2015.